

Kshitiz Bhurtel

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Experienced Software Engineer with over 3+ years of expertise in full-stack development. Skilled in React, Spring Boot, and Django, with a focus on optimizing system performance and enhancing data security. Proven ability to lead teams, develop scalable applications, and implement practical solutions to improve efficiency and reduce operational costs. Experienced with AWS, Jenkins, and OpenCV for real-time data processing. Focused on delivering impactful results through data-driven development and practical technology solutions.

Professional Experience

SOFTWARE ENGINEER II | DELOITTE | MAR 2022 - PRESENT

Backend Lead, Cybersecurity Benchmarking

- Developed a **cybersecurity benchmarking application** used by **40+ organizations** to assess **maturity levels**, identify **threats**, analyze **gaps**, and **recommend risk mitigation measures**, improving security posture by **27%**.
- Designed and implemented **authorization, RBAC, and benchmarking logic** using Django and Python, ensuring **~100%** secure access control and enhancing reporting accuracy by **35%**.
- **Led a team of 3 backend developers**, conducting weekly code reviews, which **improved code quality**.
- Optimized application scalability and performance by deploying on AWS and integrating with PostgreSQL, achieving a **15% reduction in response time**.

Software Engineer, Elevance Health

- Analyzed and optimized a monolithic Spring Boot codebase, **enhancing the Quick Links retrieval feature**, reducing runtime from **9 seconds to 4 seconds**, and improving code maintainability.
- Refactored the feature into a **separate entity**, increasing modularity and reusability, and successfully integrated it into a **new microservices-based application**.
- Identified and secured **PII/PHI data across 3 databases (~2,000 columns each)** by implementing **tokenization and encryption**, ensuring **100%** compliance with data security standards and reducing potential data exposure.

Software Engineer, Autonomous Driving Synthetic Data generation and testing

- Simulated **real-time crash scenarios** in Carla Simulation using Python scripts and tested them against the Autoware AD Stack, identifying **12%** more edge cases and improving safety validation efficiency by **5%**.
- Generated a **synthetic dataset for AD models**, creating **100K+** images with semantic and instance segmentation, reducing data collection costs by **~80%** compared to real-world data acquisition.
- Optimized bounding box mapping logic using OpenCV, reducing processing time from **20 seconds to 1 second**, which cut dataset upload time (**~50K samples**) to S3 from **20 hours to 2 hours**, increasing pipeline efficiency by **90%**.
- Trained models using the synthetic dataset, achieving **5%** improvement in accuracy compared to real-world data, demonstrating the effectiveness of simulated training data for diverse driving conditions.
- Streamlined dataset generation pipeline using **Jenkins and AWS**, reducing manual intervention by **75%**.

Contract Experience

FULL STACK DEVELOPER | BITPOSTAGE | OCT 2023 – PRESENT

- Designed and developed a **crypto-powered postage purchasing platform**, enabling users to seamlessly buy postage labels using cryptocurrencies from **USPS, UPS and FEDEX**
- The application was built in React frontend and **FastAPI backend**, deployed on **AWS** to ensure **99.9% uptime** and **scalable infrastructure** for future growth.
- **Replaced a third-party payment verification system** with a **custom-built BTC, ETH, XMR, and XRP validator**, **reducing transaction verification time by 60%** and **cutting third-party costs by 100%**.

Skills

Java • Python • JavaScript • React • Spring Boot • Django • REST APIs • PostgreSQL • MongoDB • DB2 • AWS (S3, EC2, Lambda, RDS, EKS) • Jenkins • WebSockets • OpenCV • Carla Simulation • Autoware • Microservices • Cloud Computing • Computer Vision • Payment Verification System • Software Optimization • Agile Development • Technical Leadership • Code Reviews • Performance Tuning

Education

AUG 2018 – AUG 2022

BE in Computer Science | New Horizon College of Engineering | Bangalore, India

Activities

Reading • Doodling • Exploring new tech trends